



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI KARTHIKEYAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

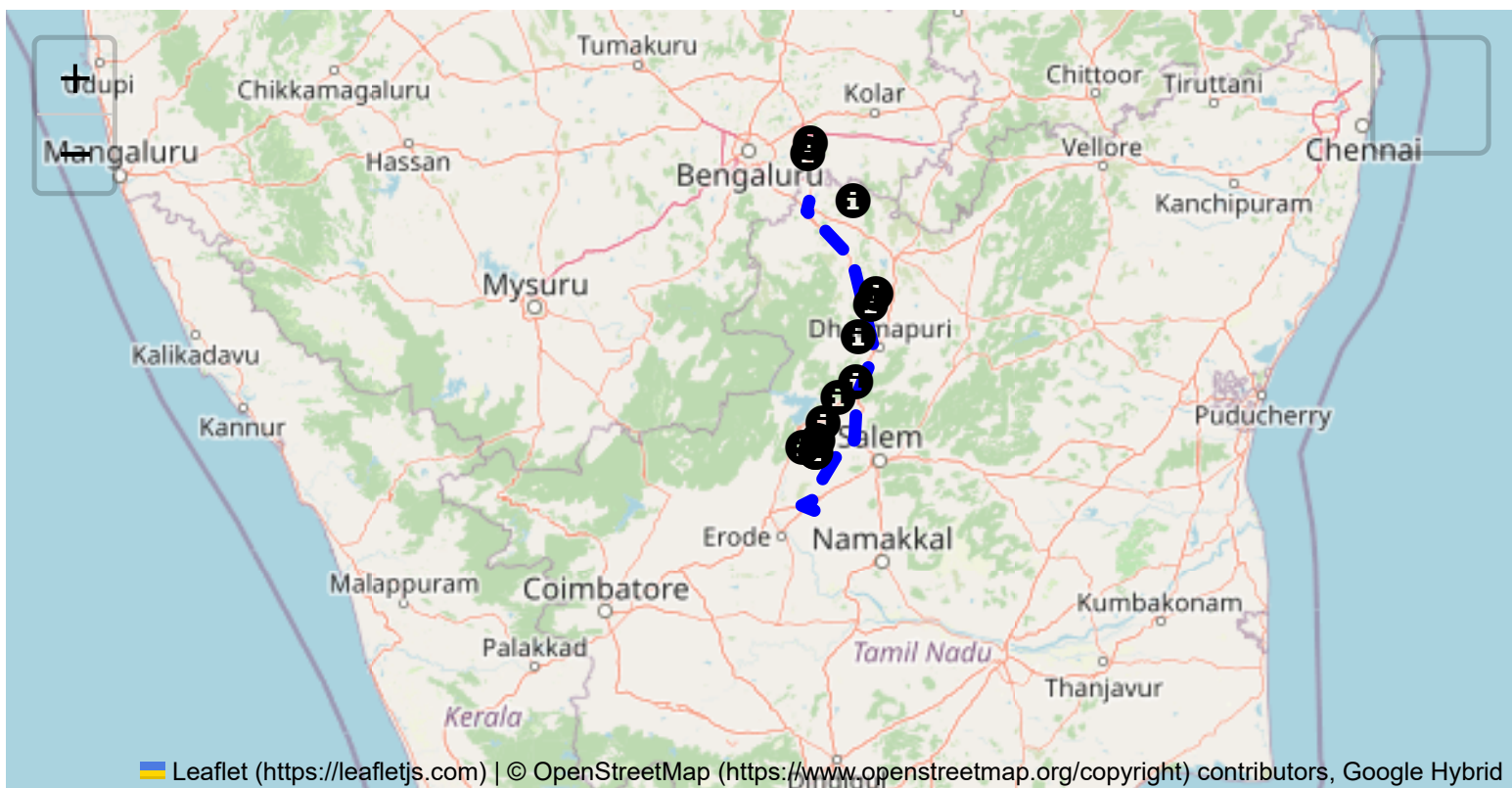
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

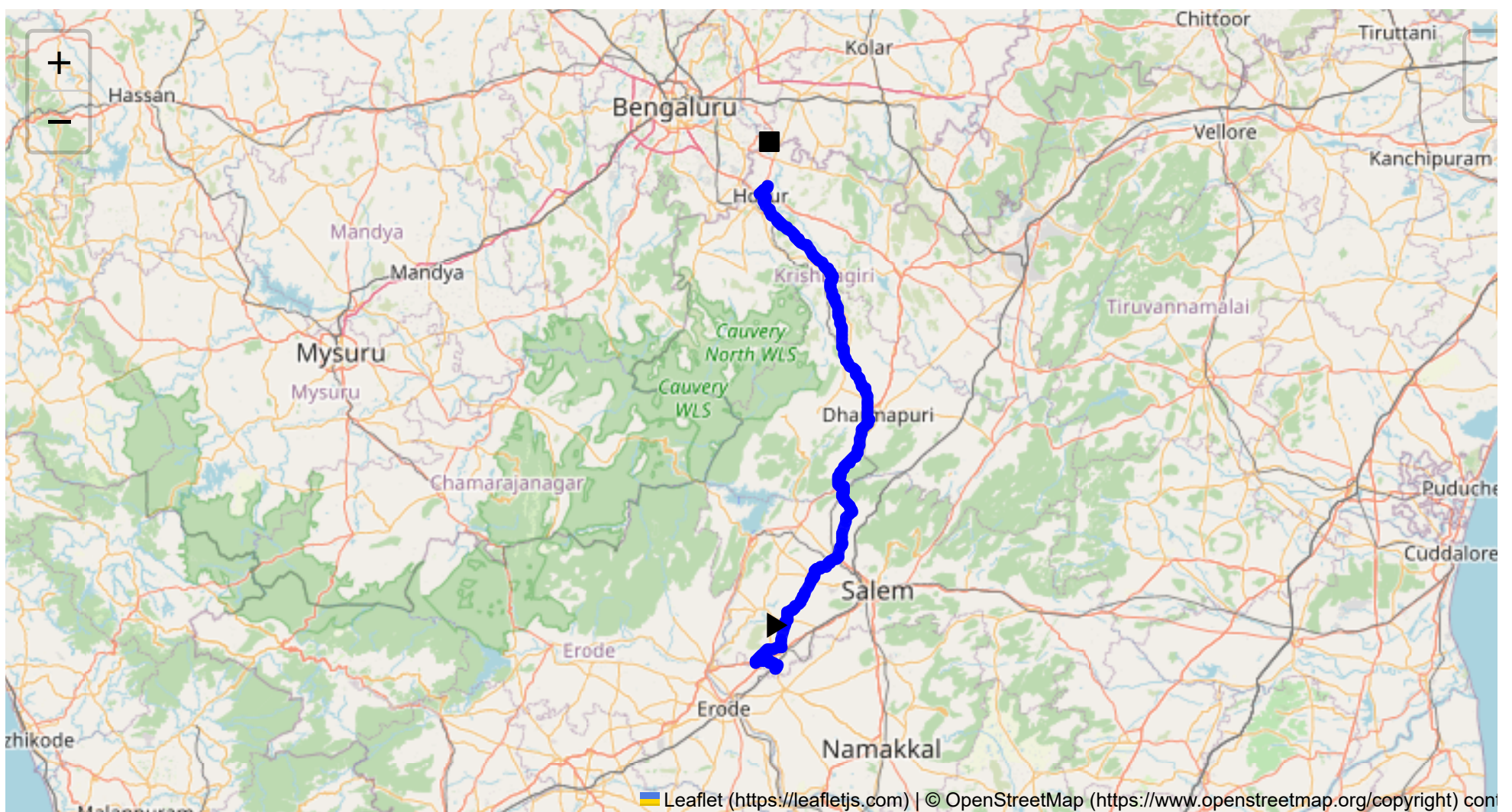
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 188.16 km
Estimated Duration: 3.9 hours
Adjusted Duration (Heavy Vehicle): 4.9 hours
Start: (11.4381, 77.8734)
End: (12.757588, 77.850128)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from CVQF+23W, Sangagiri, Tamil Nadu to QV52+22V, Avalapalli, Tamil Nadu spans approximately 188 kilometers. It passes through a mix of state highways, including the Salem – Kochi Highway and Omalur-Sankagiri Road, and regional roads connecting several towns and villages.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu generally experiences a tropical climate. The route may encounter monsoon rains from October to December, which can cause flooding and reduced visibility. Also, during April to June, high temperatures can cause road surface expansion and potential hazards due to overheating vehicles.

3. Analysis of Traffic Patterns: Traffic congestion is likely around towns such as Salem and Dharmapuri, particularly during peak hours (8-10 AM and 5-7 PM). The Bhavani Main Road and Salem - Kochi Highway are known for high traffic volumes, and weekend travel can experience increased congestion due to local commuters.

4. Assessment of Road Quality and Infrastructure: Overall road quality varies. Some sections, especially near Sangagiri and Salem, have well-maintained roads, while rural stretches may have uneven surfaces with potholes. Infrastructure, including signage and lane markings, can be sparse in remote sections.

5. Suggestions for Alternative Routes for Emergencies: In case of emergency blockages, drivers may consider detours via NH 44 for sections near Dharmapuri and Hosur, providing a more direct and often quicker route to bypass local road issues.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu mandates strict adherence to hazardous materials handling and transport regulations. Trucks must have clear labeling, comply with weight restrictions, and observe specific travel hours, primarily during off-peak times, to minimize public risk.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: The Salem and Dharmapuri segments have recorded incidents involving heavy vehicles, often due to brake failures on slopes or congestion-induced collisions. Hazardous material-specific incidents are less common but require careful adherence to protocols to prevent accidents.

8. Environmental Considerations and Sensitive Areas: The route passes through sensitive ecological zones near Salem. Drivers should be mindful of waste disposal and emissions. Noise and pollution control are crucial to prevent disturbance to local wildlife and communities.

9. Analysis of Communication Coverage: While most the route enjoys good mobile network coverage, some rural stretches near Erumaipatti and Desavilakku may experience spotty connectivity, requiring offline navigation preparation.

10. Estimated Emergency Response Times for Different Route Segments: Emergency response in urban areas like Salem is swift, typically within 20-30 minutes. In remote sections, such as between Erumaipatti and Dharmapuri, response times may extend to over an hour due to limited healthcare and emergency facilities.

11. Overall Summary of Risk Assessment: The route presents moderate risk for heavy vehicle transport, primarily due to variable road quality and potential weather-related challenges. Traffic congestion and regulatory compliance are additional considerations. Adequate preparation, including route timing adjustments, awareness of alternate routes, and proactive vehicle maintenance, will mitigate most risks. Proper training in emergency response and environmental sensitivity aligns transport practices with safety and regulatory standards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44077, 77.87533	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45352, 77.85698	30 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45545, 77.81812	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56352, 77.90078	30 KM/Hr
11	Turn	Medium	11.59189, 77.90645	30 KM/Hr
12	Turn	Medium	11.59328, 77.90809	30 KM/Hr
13	Turn	Medium	11.63867, 77.94841	30 KM/Hr
14	Turn	High	11.64217, 77.95200	15 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	Medium	11.67858, 77.96834	30 KM/Hr
17	Turn	Medium	11.69776, 77.97983	30 KM/Hr
18	Turn	Medium	11.70179, 77.97900	30 KM/Hr
19	Turn	Medium	11.71394, 77.99572	30 KM/Hr
20	Turn	Medium	11.71916, 78.01117	30 KM/Hr
21	Turn	Medium	11.72504, 78.01680	30 KM/Hr
22	Turn	Medium	11.73766, 78.03445	30 KM/Hr
23	Turn	High	11.73651, 78.03868	15 KM/Hr
24	Turn	Medium	11.74120, 78.04588	30 KM/Hr
25	Turn	Medium	11.77998, 78.05786	30 KM/Hr
26	Turn	Medium	11.86915, 78.08382	30 KM/Hr
27	Turn	Medium	11.87998, 78.08170	30 KM/Hr
28	Turn	Medium	11.91266, 78.05852	30 KM/Hr
29	Turn	Medium	11.92653, 78.06517	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	Medium	11.93196, 78.06473	30 KM/Hr
31	Turn	Medium	11.93473, 78.06268	30 KM/Hr
32	Turn	Medium	11.93735, 78.05344	30 KM/Hr
33	Turn	Medium	11.95498, 78.04689	30 KM/Hr
34	Turn	High	11.95847, 78.04989	15 KM/Hr
35	Turn	High	11.96011, 78.04944	15 KM/Hr
36	Turn	Medium	11.96251, 78.05104	30 KM/Hr
37	Turn	High	11.96427, 78.05118	15 KM/Hr
38	Turn	High	11.96526, 78.04950	15 KM/Hr
39	Turn	High	11.96637, 78.04923	15 KM/Hr
40	Turn	Medium	11.96859, 78.05102	30 KM/Hr
41	Turn	Medium	12.12010, 78.13282	30 KM/Hr
42	Blind Spot	Blind Spot	12.51381, 78.03208	10 KM/Hr
43	Turn	Medium	12.51395, 78.02897	30 KM/Hr
44	Turn	High	12.51632, 78.02571	15 KM/Hr
45	Turn	Medium	12.52916, 78.02035	30 KM/Hr
46	Turn	Medium	12.60973, 77.93664	30 KM/Hr
47	Turn	Medium	12.65345, 77.88778	30 KM/Hr
48	Turn	Medium	12.65585, 77.88617	30 KM/Hr
49	Turn	Medium	12.65767, 77.88305	30 KM/Hr
50	Turn	Medium	12.69040, 77.85486	30 KM/Hr
51	Turn	Medium	12.69448, 77.85412	30 KM/Hr
52	Turn	Medium	12.71737, 77.83927	30 KM/Hr
53	Turn	Medium	12.71956, 77.83555	30 KM/Hr
54	Turn	High	12.71961, 77.83551	15 KM/Hr
55	Turn	Medium	12.72411, 77.84014	30 KM/Hr
56	Turn	Medium	12.72579, 77.84074	30 KM/Hr
57	Turn	Medium	12.72679, 77.84242	30 KM/Hr
58	Turn	High	12.72829, 77.84300	15 KM/Hr
59	Turn	Medium	12.73474, 77.83019	30 KM/Hr
60	Turn	High	12.73549, 77.82949	15 KM/Hr
61	Turn	Medium	12.73607, 77.82962	30 KM/Hr
62	Turn	Medium	12.75648, 77.84984	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
27	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
31	pharmacy	H2 Pharmacy	12.7301802, 77.8401037	30 km/h	Medium
32	clinic	Meera Nursing Home, Krishnagiri	12.7326185, 77.8292795	30 km/h	Medium
35	hospital	Amma Hospital	12.7393554, 77.8326981	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
23	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
29	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
30	school	Panchayat Union Primary School	12.6835154, 77.8604798	30 km/h	Medium
33	school	Vijay Vidyalaya School	12.7397242, 77.8308427	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	school	Maharishi Vidhya Mandir Primary School	12.7397254, 77.8324166	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44077, 77.87533



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56352, 77.90078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64712, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69776, 77.97983



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70179, 77.97900



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73766, 78.03445



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77998, 78.05786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86915, 78.08382



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87998, 78.08170



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91266, 78.05852



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92653, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93196, 78.06473



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93473, 78.06268



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93735, 78.05344



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95498, 78.04689



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95847, 78.04989



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96011, 78.04944



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96251, 78.05104



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96427, 78.05118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96526, 78.04950



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96637, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96859, 78.05102



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12010, 78.13282



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51381, 78.03208



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51395, 78.02897



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51632, 78.02571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52916, 78.02035



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60973, 77.93664



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.65767, 77.88305



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.69040, 77.85486



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.69448, 77.85412



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71737, 77.83927



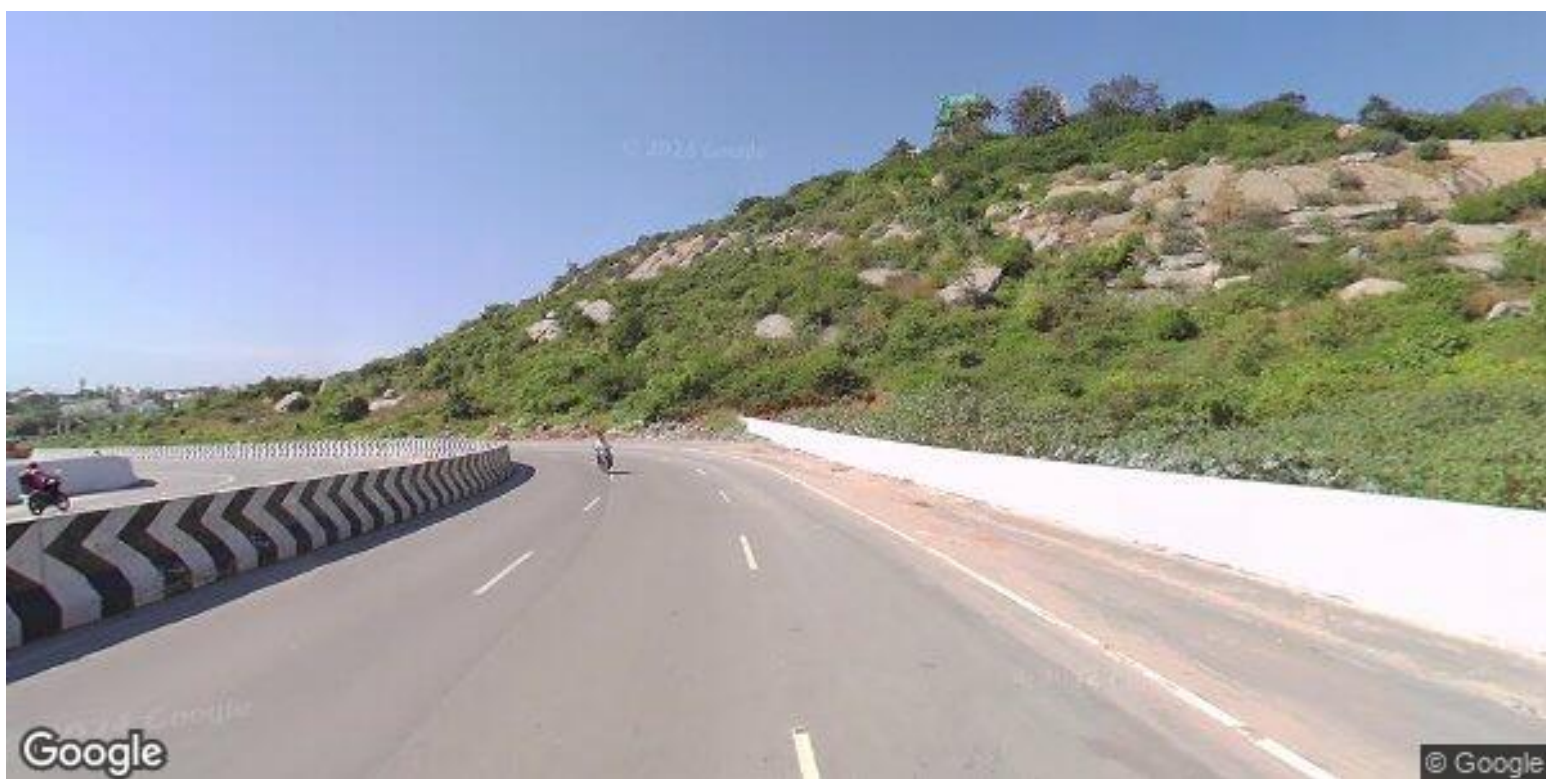
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71956, 77.83555



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.71961, 77.83551



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72411, 77.84014



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72579, 77.84074



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72679, 77.84242



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.72829, 77.84300



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.73474, 77.83019



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.73549, 77.82949



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.73607, 77.82962



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.75648, 77.84984
